分布式系统（英文）

课程名称/Course Title： 分布式系统（英文）/ Principles of Distributed Systems

课程代码/Course Code：

任课教师/Instructor(s)： 法赫德萨巴赫/ Fahad Sabah

开课学部（院）/Faculty： 计算机学院

1．课程概要/Course Summary				
课程名称（中文） Course Title (Chinese)	分布式系统（英文）			
课程名称（英文） Course Title (English)	Principles of Distributed Systems			
适用层次(可多选) Applicable level(s)	☑ 学硕 (Academic master)	☑ 学博 (Academic Ph.D)	☐专硕 (Professional master)	☐专博 (Professional Ph.D)
授课语言 Teaching Language	英语 English	适用学科/专业学位类别（领域） Discipline	软件工程 Software Engineering	
学分数 Course Credit(s)	48	开课学期 Semester		
总学时 Teaching Hours in Total	共 48 学时 48 teaching hours	实验/实践学时 Hours for Experiments /Practice	共 0 学时 0 teaching hours	
预修课程要求 Pre-requisite Course(s)	编程基础、操作系统、计算机网络 Programming Fundamentals, Operating Systems, Computer Networks			
课程简介 Course Introduction （150-300 字）	本研究生课程为期 12 周，每周 4 学时。在云计算、去中心化应用和全球数字基础设施的时代，分布式系统构成了现代技术的核心从实时金融交易和物联网网络，到 AI 驱动平台和区块链生态系统。课程深入探讨分布式系统的复杂设计、理论基础和实际实现，培养学生解决可扩展性、容错性和安全性挑战的能力。课程结合严谨的学术研究与动手实践，涵盖基础原理（共识算法、一致性模型、CAP 定理）与前沿技术（无服务器架构、抗量子密码学、分布式账本）。学生将使用 Apache Kafka、Kubernetes 和 Tendermint 等行业标准开源工具，构建实时分析管道、拜占庭容错网络和符合 GDPR 的数据库等弹性系统。研究是课程的核心：学员将剖析经典论文、复现实验，并为开源项目贡献代码，促进与全球开发者社区的协作。课程面向有志成为研究人员、架构师和工程师的学生，最终以顶点项目作结学员需针对未解决问题（如量子-经典混合共识或边缘计算优化）设计创新解决方案。 Spanning 12 weeks with four hours of student engagement per week, this offering			

	sits in an era shaped by cloud platforms, decentralized services, and planetary-scale infrastructure context in which distributed systems quietly underpin everyday reliability: payments and streaming pipelines, device fleets and edge caches, large-model serving and replicated control planes. This graduate-level course examines how such systems are designed, reasoned about, and built, so you can navigate scalability, fault tolerance, and security-aware design in environments where partial failure and ambiguous timeouts are normal, not exceptional. The curriculum joins rigorous reading with sustained hands-on work: foundational ideas models of failure and time, consistency, replication, quorums, CAP, and consensus meet contemporary themes from the reading list, including eventual consistency, CRDTs, transactions and logs, stream-oriented processing, and observability. This course have practical activities (a sequenced set of programming assignments from RPC measurement through replication, quorums, and Raft analysis using stacks such as gRPC and optional fault-injection tooling like Toxiproxy in local or container testbeds). Research-style engagement is central: you will study seminal and influential papers, relate them to real systems, and, where exercises demand it, reproduce assigned behaviors or measurements. The course culminates in an individual capstone: a distributed-systems project with fault or partition experiments, evaluation, and a research-style technical report preparing aspiring researchers, architects, and engineers to contribute responsibly to the next generation of distributed systems in academia, industry, and beyond.

2. 授课团队/Teaching group

姓名 Name	法赫德萨巴赫 Fahad Sabah	职称 Title		工作证号 ID	
姓名 Name		职称 Title		工作证号 ID	
姓名 Name		职称 Title		工作证号 ID	

3. 教学目标/Course Objective（100-200 字）

这门研究生课程培养学生掌握设计和创新分布式系统所需的先进理论与实用技能，适用于学术和工业领域。课程将严谨的理论框架与开源工具实践相结合，使学生精通共识算法（如 Paxos、Raft、BFT）、一致性模型（线性一致性、因果一致性）和容错机制等核心概念，并深入理解 CAP 定理所揭示的系统设计权衡。通过业界标准工具（如 Apache Kafka、Kubernetes、CockroachDB）和形式化验证工具 TLA+，学生将构建可扩展、高可用的系统，包括流处理管道、去中心化数据库和拜占庭容错网络。课程特别关注安全与隐私，涵盖零知识证明（zk-SNARKs）和符合 GDPR 的架构设计，确保系统能应对技术和法规的双重挑战。课程还探索无服务器计算、边缘编排和抗量子密码学等新兴范式，以提升学生的前瞻能力。研究是课程的核心组成部分：学生将精读经典论文、进行混沌工程实验，并参与 Apache Flink、Hyperledger 等开源项目贡献。最后的顶点项目要求学生开发创新解决方案（如优化广域网共识协议或设计量子-经典混合系统），并以学术会议形式展示成果。通过促进与开源社区的合作并紧跟前沿研究，本课程为学生将来在学术界、研发实验室或云原生行业担任领导角色做好准备，使其能够以创新和精准的方式应对全球规模的分布式系统挑战。

By the end of this graduate course, students will be able to explain and apply core concepts of distributed systems including failure and timing models, message delivery semantics, consistency and replication, quorums, the CAP trade-off, and consensus using assigned texts (Designing Data-Intensive Applications; Distributed

Systems: Principles and Paradigms) and weekly research and practitioner readings. They will analyze protocols and system designs from the literature, connect informal operational behavior to explicit assumptions, and evaluate designs under faults and partitions. Through take-home practical activities, they will implement and measure representative mechanisms (e.g., RPC behavior, logical time, idempotent APIs, hashing and replication patterns, quorum behavior, Raft-related analysis) with tests and optional fault injection. They will complete a midterm that assesses conceptual mastery through the consensus unit, and an individual capstone implementation or analysis as approved reported as a research-style technical paper with reproducible experiments, failure scenarios, and clear citations. Overall, graduates should be prepared to reason precisely about partial failure and trade-offs, and to communicate design decisions defensively in both academic and engineering settings.

4．教学内容及进度安排/Course Content & Schedule

课次 No.	教学周 Week	教学内容/Contents	作业/实验 Assignment
1	1	• 分布式系统导论：分布式系统的定义、目标及类型 • Introduction to Distributed Systems: Definition, Goals, and Types of Distributed Systems	关于现实世界的分布式系统（例如云服务、点对点系统）的演讲。讨论日常生活中使用的例子。 Presentation on real-world distributed systems (e.g., cloud services, peer-to-peer systems). Discussion of examples used in daily life.
2	2	• 分布式系统架构：架构风格（集中式、分散式、混合式）、中间件与自我管理 • Distributed System Architectures: Architectural Styles (Centralized, Decentralized, Hybrid), Middleware and Self-Management	设计一个简单的客户端-服务器架构，并使用套接字编程实现服务器-客户端通信。 Design a simple client-server architecture and implement server-client communication using socket programming.
3	3	• 分布式系统中的进程：线程、虚拟化以及客户端和服务器的角色 • Processes in Distributed Systems: Threads, Virtualization, and Role of Clients and Servers	构建一个多线程的客户端-服务器应用，并设置虚拟化（例如虚拟机和网络）。 Build a multi-threaded client-server application and set up virtualization (e.g., VMs and networking).
4	4	• 通信基础：RPC 与面向消息通信 • Communication Fundamentals: RPC and Message-Oriented Communication	用 Python/Java 实现 RPC，并探索面向消息的通信工具（RabbitMQ/Kafka）。 Implement RPC using Python/Java and explore message-oriented communication tools (RabbitMQ/Kafka).
5	5	• 分布式系统中的命名：平面命名、结构化命名和 DHT	用 DHT 实现一个简单的命名系统，并设计一个基本的 DNS 类系

		• Naming in Distributed Systems: Flat Naming, Structured Naming, and DHT	统。 Implement a simple naming system using DHT, and design a basic DNS-like system.
6	6	• 分布式系统中的同步：时钟同步、互斥、选举算法 • Synchronization in Distributed Systems: Clock Synchronization, Mutual Exclusion, Election Algorithms	实现 Lamport 的逻辑时钟或向量时钟以及基本的选举算法（例如 Bully 或 Ring）。 Implement Lamport's Logical Clocks or Vector Clocks and basic election algorithms (e.g., Bully or Ring).
7	7	• 一致性与复制：以数据为中心与以客户端为中心的一致性模型，副本管理 • Consistency and Replication: Data-Centric vs. Client-Centric Consistency Models, Replica Management	实现具有一致性保证的复制，并设计缓存机制，以平衡一致性与性能之间的权衡。 Implement replication with consistency guarantees and design a caching mechanism for trade-offs between consistency and performance.
8	8	• 分布式系统中的容错：故障模型、恢复机制与可靠通信 • Fault Tolerance in Distributed Systems: Failure Models, Recovery Mechanisms, and Reliable Communication	实施容错通信协议，并为系统创建恢复机制。 Implement fault-tolerant communication protocols and create a recovery mechanism for systems.
9	9	• 分布式系统中的安全性：密码学、认证与访问控制 • Security in Distributed Systems: Cryptography, Authentication, and Access Control	实现使用 TLS/SSL 或基础密码学（如 RSA）的安全通信。建立安全的客户端-服务器通信并实现认证。 Implement secure communication using TLS/SSL or basic cryptography (e.g., RSA). Set up secure client-server communication with authentication.
10	10	• 分布式基于对象的系统：Java RMI，CORBA • Distributed Object-Based Systems: Java RMI, CORBA	使用 Java RMI 或 CORBA 实现分布式对象系统。探索关键操作，如对象引用和方法调用。 Implement a distributed object system using Java RMI or CORBA. Explore key operations like object references and method invocation.
11	11	• 分布式文件系统：NFS、Coda、GFS • Distributed File Systems: NFS, Coda, GFS	实现一个基本的分布式文件共享系统。构建并模拟一个简单的文件锁定机制。

			Implement a basic distributed file-sharing system. Build and simulate a simple file-locking mechanism.
12	12	• 基于网络的分布式系统与综述：网络服务、网络系统的容错性与网页托管的复制 • Web-Based Distributed Systems and Review: Web Services, Fault Tolerance in Web Systems, and Replication for Web Hosting	用 REST API/SOAP 实现一个 Web 服务。呈现聚焦实施和成果的期末项目。 Implement a web service using REST APIs/SOAP. Present the final project focusing on implementation and results.

5. 课程考核及成绩评定/Course Assessment & Grading

考核指标* Assessment Criteria	权重 Percentage	评定标准 Assessment Standard
出勤 Attendance		
课堂表现 Participation		
作业/实验 Assignment(s)	30	按时提交所有作业 Timely submission of all assignments 代码符合规范要求 Code meets specified standards 功能实现完整 Complete implementation of required functions 报告内容清晰准确 Clear and accurate reports
课程论文 Course Paper		
课程考试 Course Exam(s).		
项目 Project(s)	70	项目完整性（40%）：实现所有要求功能 Completeness (40%): All required features implemented 代码质量（30%）：符合 PEP8 规范，结构清晰 Code quality (30%): PEP8 compliant, well-structured 文档（20%）：包含清晰的 README 和注释 Documentation (20%): Clear README and comments 创新性（10%）：体现创造性解决方案 Creativity (10%): Demonstrates innovative solutions

* 各项考核指标可自由设置，总权重为 100%。

6. 教材/Textbook(s) （如使用自编讲义，请在“名称”列中备注说明）

序号 No.	名称 Title	作者 Author(s)	标准书号 ISBN	出版机构 Publisher	出版日期 Publication Date	是否必读 Mandatory or Elective

1	Distributed Systems: Principles and Paradigms	Andrew S. Tanenbaum, Maarten van Steen	978-1530281756	CreateSpace Independent	2016	Mandatory
2	Introduction to Reliable and Secure Distributed Programming	Christian Cachin, Rachid Guerraoui, Luís Rodrigues	978-3642152597	Springer	2011	Elective
7.教学参考资料/Reading Materials and References						
序号 No.	名称 Title	作者 Author(s)	标准号码 ISBN/DIO	出版机构 Publisher	出版日期 Publication Date	是否必读 Mandatory or Elective
1	无					